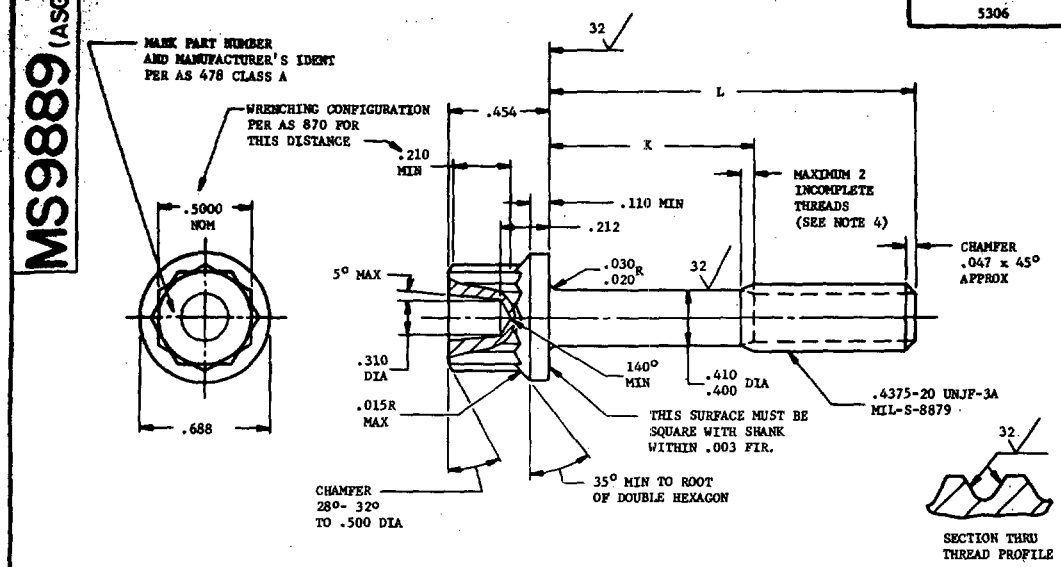
# MS9889 (ASG)

FED. SUP CLASS 5306




| PART NO. | L | K | APPROX. WEIGHT LB/100 | PART NO. | L | K | APPROX. WEIGHT LB/100 | PART NO. | L | K | APPROX. WEIGHT LB/100 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| MS9889-04 | .688 | .105- .125 | 5.58 | MS9889-22 | 1.812 | .627- .687 | 9.71 | MS9889-40 | 3.875 | 2.690-2.750 | 17.30 |
| MS9889-05 | .750 | .105- .125 | 5.80 | MS9889-23 | 1.875 | .690- .750 | 9.94 | MS9889-41 | 4.000 | 2.815-2.875 | 17.76 |
| MS9889-06 | .812 | .105- .125 | 6.03 | MS9889-24 | 1.938 | .752- .812 | 10.17 | MS9889-42 | 4.125 | 2.940-3.000 | 18.22 |
| MS9889-07 | .875 | .105- .125 | 6.26 | MS9889-25 | 2.000 | .815- .875 | 10.40 | MS9889-43 | 4.250 | 3.065-3.125 | 18.68 |
| MS9889-08 | .938 | .105- .125 | 6.49 | MS9889-26 | 2.125 | .940-1.000 | 10.86 | MS9889-44 | 4.375 | 3.190-3.250 | 19.14 |
| MS9889-09 | 1.000 | .105- .125 | 6.72 | MS9889-27 | 2.250 | 1.065-1.125 | 11.32 | MS9889-45 | 4.500 | 3.315-3.375 | 19.60 |
| MS9889-10 | 1.062 | .105- .125 | 6.95 | MS9889-28 | 2.375 | 1.190-1.250 | 11.78 | MS9889-46 | 4.625 | 3.440-3.500 | 20.06 |
| MS9889-11 | 1.125 | .105- .125 | 7.18 | MS9889-29 | 2.500 | 1.315-1.375 | 12.24 | MS9889-47 | 4.750 | 3.565-3.625 | 20.51 |
| MS9889-12 | 1.188 | .105- .125 | 7.41 | MS9889-30 | 2.625 | 1.440-1.500 | 12.70 | MS9889-48 | 4.875 | 3.690-3.750 | 20.98 |
| MS9889-13 | 1.250 | .105- .125 | 7.64 | MS9889-31 | 2.750 | 1.565-1.625 | 13.16 | MS9889-49 | 5.000 | 3.815-3.875 | 21.43 |
| MS9889-14 | 1.312 | .128- .188 | 7.87 | MS9889-32 | 2.875 | 1.690-1.750 | 13.62 | MS9889-50 | 5.125 | 3.940-4.000 | 21.89 |
| MS9889-15 | 1.375 | .190- .250 | 8.10 | MS9889-33 | 3.000 | 1.815-1.875 | 14.08 | MS9889-51 | 5.250 | 4.065-4.125 | 22.34 |
| MS9889-16 | 1.438 | .252- .312 | 8.33 | MS9889-34 | 3.125 | 1.940-2.000 | 14.54 | MS9889-52 | 5.375 | 4.190-4.250 | 22.79 |
| MS9889-17 | 1.500 | .315- .375 | 8.56 | MS9889-35 | 3.250 | 2.065-2.125 | 15.00 | MS9889-53 | 5.500 | 4.315-4.375 | 23.23 |
| MS9889-18 | 1.562 | .378- .438 | 8.79 | MS9889-36 | 3.375 | 2.190-2.250 | 15.46 | MS9889-54 | 5.625 | 4.440-4.500 | 23.67 |
| MS9889-19 | 1.625 | .440- .500 | 9.02 | MS9889-37 | 3.500 | 2.315-2.375 | 15.92 | MS9889-55 | 5.750 | 4.565-4.625 | 24.12 |
| MS9889-20 | 1.688 | .502- .562 | 9.25 | MS9889-38 | 3.625 | 2.440-2.500 | 16.38 | MS9889-56 | 5.875 | 4.690-4.750 | 24.56 |
| MS9889-21 | 1.750 | .565- .625 | 9.48 | MS9889-39 | 3.750 | 2.565-2.625 | 16.84 | MS9889-57 | 6.000 | 4.815-4.875 | 25.01 |

1. SHANK SHALL BE STRAIGHT WITHIN .0025 PER INCH OF BOLT LENGTH.
2. THE CONCENTRICITY OF THREAD PD IN RELATION TO THE SHANK SHALL BE WITHIN .006 FIR.
3. THE CONCENTRICITY OF THE SHANK IN RELATION TO THE WASHER FACE DIAMETER AND DOUBLE HEXAGON OD SHALL BE WITHIN .010 FIR.
4. INCOMPLETE THREADS NOT TO ENTER FILLET.
5. MATERIAL: CORROSION RESISTANT STEEL AMS 5616.
6. MANUFACTURING SPECIFICATION: AMS 7470 EXCEPT HEAD SHALL BE UPSET.
7. HEAD TO SHANK FILLET SHALL BE COLD ROLLED AFTER HEAT TREATMENT TO REMOVE ALL VISUAL EVIDENCE OF GRINDING OR TOOL MARKS.
8. HARDNESS: ROCKWELL C32-38.
9. SURFACE TEXTURE: USAS B46.1-1962. UNLESS OTHERWISE SPECIFIED, SURFACES TO BE 125 MICROINCHES EXCEPT UPSET HEAD.
10. MAGNETIC PARTICLE INSPECTION PER AMS 2640.
11. BREAK SHARP EDGES .003-.015 UNLESS OTHERWISE SPECIFIED.
12. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: LINEAR DIMENSIONS ± .010, ANGULAR DIMENSIONS ± 5°.
13. DO NOT USE UNASSIGNED PART NUMBERS.

AS & AMS ARE SOCIETY OF AUTOMOTIVE ENGINEERS, INC. PUBLICATIONS.
THIS STANDARD WAS DEVELOPED COOPERATIVELY WITH THE MILITARY SERVICES BY THE SAE AEROSPACE PART STANDARDS DIVISION

APPROVED 3 Mar 69 REVISED

This military standard is approved by the Department of the Air Force and the Naval Air Systems Command and is mandatory for use by these activities. All other military activities are required to employ this standard where suitable.

Review activities: USAF - 82,85

| P.A. | TITLE | MILITARY STANDARD |
|---|---|---|
| Other Cust: USAF -11; Navy - AS | BOLT, MACHINE - DOUBLE HEXAGON EXTENDED WASHER HEAD, AMS 5616, .4375-20 UNJF-3A | MS9889 (ASG) |
| PROCUREMENT SPECIFICATION | SUPERSEDES: | SHEET 1 OF 1 |

DD FORM SEP 67 672-1 (Limited coordination)
ASG use only

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

Project No. 5306-F039

☆ U. S. GOVERNMENT PRINTING OFFICE: 1969-341-517/A-5040

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:27Z
Check the source to verify that this is the current version before use.